File-Level Is Not Symbol-Level: What Code Graphs Miss and the Issue Already States

300 tasks, then 2,232: the granularity gap is invariant, the graph is worth two points at symbol level and none at file level, and a three-point hand-written rule was really training data

Abstract

Repository-level agents are judged on whether they resolve an issue; the retrieval that feeds them is reported at file granularity. We measure the granularity an agent edits. On 300 SWE-bench Lite tasks we extract the **354 symbols the reference patches actually touch**: the same ranking that recovers **62.4%** of gold files at rank 10 recovers **27.5%** of gold symbols, a 2.3× gap that is 2.28× again on 2,232 tasks. **Eight ways of adding a code graph leave symbol-level recall unchanged** on 300 tasks and buy 2.2 points on 2,232 — none of it at file level. What pays is what the issue states: a ranker over sixteen cheap features reaches **36.1%** of symbols and **64.3%** of files, and an evidence rule that looked worth 2.9 points was a ranker short of training data. We release the benchmark and the harness.

1. Introduction

An agent that cannot see a repository must search it: the context holds a few hundred symbols, the repository tens of thousands, and everything downstream depends on the right one being among the few.

LocAgent reports 92.7% file-level accuracy on SWE-bench-style tasks [2] — but a large module holds dozens of definitions, and opening the right file still leaves the agent to find the one function the fix belongs in.

We measure the fine granularity. For each of 300 SWE-bench Lite tasks the reference patch gives the symbols the fix touched, and the same ranking that finds 62.4% of gold files at rank 10 finds 27.5% of gold symbols. **The gap is 2.3×, and it is not an artefact — it is one list, scored two ways.**

One instance: `matplotlib__matplotlib-24970` reports NumPy 1.24 deprecation warnings and its traceback prints file and line, so the file is never in doubt — every arm puts `lib/matplotlib/colors.py` in its top five. Not one puts `Colormap.__call__` in its top ten, though the 200-candidate pool contains it.

Second, we test the standard remedy: build a call graph and let structure carry the rest. None of eight arms improves symbol-level recall at ten slots — not even personalised PageRank — and the most aggressive costs 4.8 points of file-level recall ($p = 0.004$). What does gain is the source the issue quotes (§5).

The explanation is arithmetic. One-hop balls reach a gold symbol in 13% of tasks but are 0.42% gold against 3.1% per text slot. And 54% of the call sites that would carry the hop resolve to no definite target.

2. Related Work

Repository-level benchmarks. SWE-bench and its Lite and Verified subsets score whether a patch makes the failing tests pass [1], and say nothing about whether the code could have been found. Our targets derive from SWE-bench Lite's patches, but symbol-level retrieval is orthogonal to resolved rate and reported as such.

Agents over repositories. SWE-agent fixes the interface and lets the model act [5]; LocAgent navigates with graph structure [2]; ARISE builds a repository graph and lifts Function R@1 from 0.43 to 0.60 [3]. These systems co-design retrieval with a model, so their numbers are not comparable to ours: the graph's marginal value is the only thing that varies.

Code graphs. PyCG reports 99.2% precision and 69.9% recall on its own benchmark [4]; CodexGraph turns a repository into a queryable code graph [6], and RepoAtlas maintains evolving repository views for agents [7]. We report the resolution rate of ours — 46% of call sites definite — and the ceiling that bounds one-hop expansion.

3. Setup

Tasks. All 300 instances of SWE-bench Lite, 12 repositories, dominated by django (114) and sympy (77). We fetch each base commit as a tarball (297 distinct commits, 2.5 GB). Nothing is executed: the measurement is over source.

Gold symbols. For each task we parse the reference patch and map every changed line to the innermost definition containing it in the *pre-fix* tree: a deleted line anchors itself, an added line the line before it, since the added text does not exist before the fix. That gives 354 definitions over 290 tasks; the other 10 are dropped: 2 change only module-level

lines, 8 only add definitions. We record rather than drop the 87 changed lines outside any definition, so hard cases cannot inflate the score. A symbol is `<module>.<qualname>`, module being the file path without `.py`; file-level metrics group the same ranking by file. A second implementation — each definition's own text, before and after the patch — reproduces 279 of 300 keys; rescoring with it moves recall by at most 0.7 points.

Query and budget. The issue text as given: no rewrite, no file hints, no LLM preprocessing — the weakest defensible query, so that no improvement can be attributed to query engineering. Every arm fills exactly 10 slots, and we report the characters those ten symbols would add to a prompt, so an arm cannot win by retrieving more.

4. Method

The graph builder. We build a definition-and-reference graph from source with Python's `ast`, emitting the same JSON shape the competition's released assets use, so one metric applies to both. Nodes are modules, classes, functions and methods, each carrying its signature, docstring, and source span; edges come in four types — `defines`, `imports`, `inherits`, `calls`.

Resolution states. What matters is not the edge list but what each edge means. We record seven states and never collapse them:

state	meaning
`local`	target defined in the same module
`import`	target reached through an import alias
`unique`	leaf name occurs exactly once in the repository
`suffix`	target matched a dotted suffix (src/ layouts)
`external`	stdlib or third-party; not a defect
`ambiguous`	more than 64 same-named candidates; refused rather than guessed
`unresolved`	no target found

On a django snapshot (2,715 files, 38,651 nodes, 209,502 edges, 10.8 s) the distribution is 34.1% unresolved, 19.2% local, 15.6% import, 15.0% ambiguous, 11.2% unique, 4.9% external. **Only 46% of call sites resolve to a definite in-repository target** — the quality of the artefact the arms expand, and the reason `graph_fuse_precise` exists.

The arms. Eight, all paired by task; `text` is BM25 over a per-symbol document (path, qualified name, signature, docstring), and the other seven fold in the graph. `graph_rerank` promotes anything one hop from the top five seeds without changing the pool, so it cannot add a candidate. `graph_open` keeps the top five and fills slots 6–10 with their neighbours from the whole graph — the only arm that can introduce a symbol BM25 never ranked. `graph_fuse` lets a candidate inherit $\lambda = 0.5$ of its best adjacent seed's score, **fixed before the run and never tuned**; `graph_fuse_precise` fuses over only the definite 46%. `graph_ppr` is personalised PageRank over the whole graph (restart 0.15, the convention, not tuned); `graph_ppr_text` adds half the diffusion mass to normalised BM25. Neighbours are unioned over `calls` and `inherits` and drawn round-robin so a hub cannot absorb the shortlist.

Two more rankings. The evidence arm reads what the issue states: a traceback frame resolves through the same symbol table we built, a named path is a suffix match, and a quoted line is found by indexing the candidate's *body* — text our symbol documents never contain. The learned arm puts sixteen signals into one ranker (§5).

Statistics. Per-task differences, so arms are compared on the same issue and repository: mean difference with a bootstrap 95% interval (2,000 resamples, fixed seed) plus an exact sign test, which decides, because ties dominate.

5. Results

The granularity gap is the headline. Table 1, one ranking, two ways of scoring it. All 354 gold symbols resolve into the graph, so the denominator is not in question.

***Table 1** — symbol vs file granularity (n = 290 scored tasks)*

granularity	recall@1	recall@5	recall@10
symbol	0.091	0.214	**0.275**
file	—	0.548	**0.624**

Line granularity agrees: the ranking covers **23.9%** of the 1,010 changed lines, so the gap is not a matching artefact. **At ten slots, no graph arm improves symbol-level recall** — Table 2 and Figure 1.

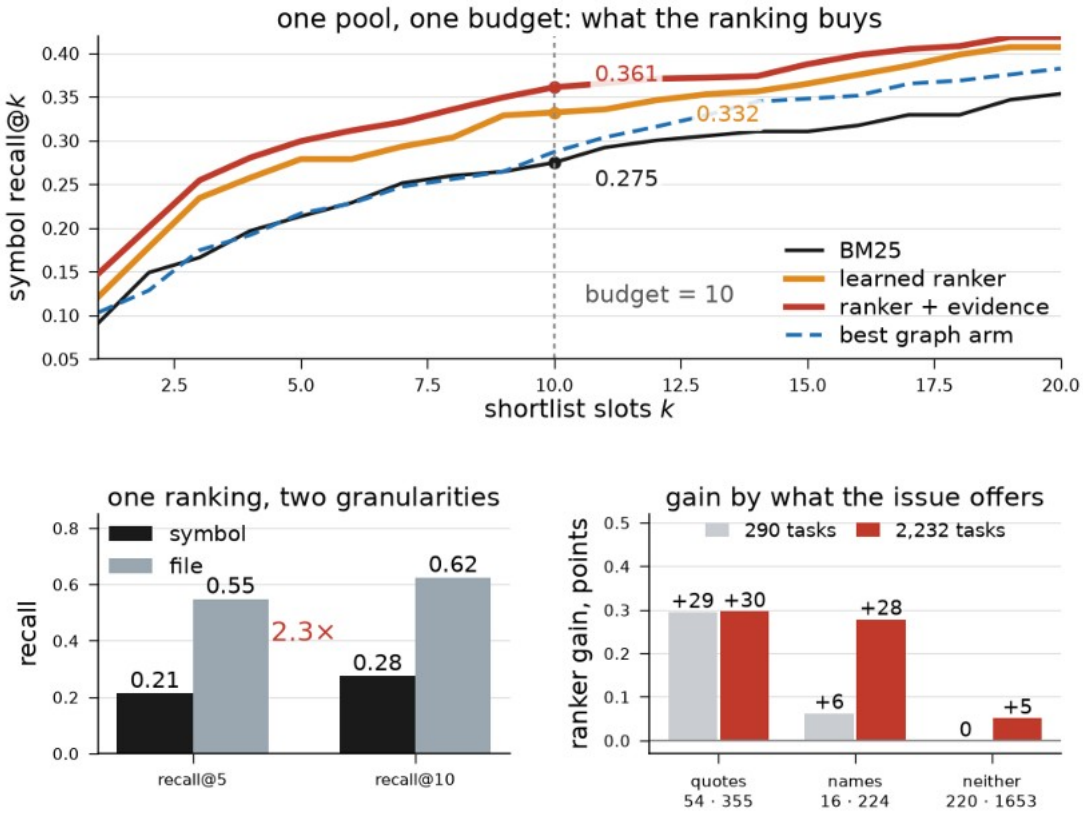


Figure 1 *— top: symbol recall@k for BM25, the best graph arm, the learned ranker and the ranker with evidence. Bottom left: the same ranking by file and by symbol. Bottom right: the ranker's gain over BM25 by what the issue offers, at both scales.*

Table 2 *— the eight arms of §4, paired, equal pool and equal character budget*

arm	r@1	r@5	r@10	file r@10	Δr@10 (95% CI)	p
text	0.091	0.214	0.275	0.624	—	—
graph_rerank	0.103	0.217	0.287	0.614	+0.012 (−0.026, 0.050)	0.50
graph_open, 1 hop	0.091	0.214	0.257	0.576	−0.018 (−0.051, 0.013)	0.46
graph_open, 2 hops	0.091	0.214	0.249	0.579	−0.026 (−0.057, 0.003)	0.18
graph_fuse	0.091	0.217	0.272	0.624	−0.003 (−0.010, 0.000)	1.00
graph_fuse_precise	0.091	0.217	0.275	0.624	0.000	1.00
graph_ppr	**0.106**	0.214	0.268	0.579	−0.007 (−0.040, 0.027)	0.72
graph_ppr_text	0.091	0.217	0.272	0.624	−0.003 (−0.010, 0.000)	1.00

The second-hop variant has the most reach (35.7%, Table 3) and is still negative. Reranking survives a wider budget: +2.9 at k = 20 (13 wins, 3 losses, p = 0.021), +0.8 at k = 50.

Diffusion is a lateral move; expansion is worse. Personalised PageRank is the only arm that improves the first slot (0.106 against 0.091), and it gains 15 gold symbols while losing 17; at file level it loses 4.5 points (p = 0.019). `graph_open` trades slots 6–10 for neighbours: 18 tasks lose a gold file, 4 gain one, −4.8 points (p = 0.004). **Reach bounds the arm; the exchange rate eats the bound** — Table 3.

Table 3 *— diagnostics, ceilings and the ablation grid (n = 290)*

quantity	value	meaning
reach, 2 hops, 1 / 3 / 5 / 10 seeds	0.191 / 0.310 / 0.357 / 0.395	radius buys reach quickly
one-hop ball, median size / gold density	21 nodes / 0.42%	41 gold symbols among 9,855 neighbours
diffusion's conversion of the ball	18 / 41 (44%)	reachable gold placed in its top 10
oracle@10, pool only / pool plus neighbours	0.600 / 0.630	a perfect reranker's ceiling, with and without expansion
tasks whose gold symbols never enter the top-500 pool	80 / 290 (27.6%)	the hard tail

One-hop expansion has an arithmetic ceiling: 38 of 290 tasks (13.1%; 41 of 354 symbols) hold a gold symbol one hop from a top-five seed, so perfect conversion is worth 11.9 points; two hops lift the ceiling to 35.7%. Conversion is not the failure — diffusion places 18 of the 41 reachable symbols in its top ten (44%) and still lands below text, because 15 gained cost 17 lost. The rate is the finding: 9,855 neighbours hold 41 answers, 0.42%, against 3.1% gold per text slot.

Where the gap is. Ten slots are a budget: recall@10 ≈ P(gold file selected) × E[recovered | depth], and the product stays near 0.25 whether the slots go to one file or ten. Given the *right* file, ten slots recover **66.5%** of gold symbols.

A learned ranker. Sixteen features — lexical score and rank, body-quote overlap, path and frame matches, four graph statistics, size and shape terms — feed a LightGBM ranker trained on six repositories, tested on the other six, three seeds averaged, nothing tuned on the test fold.

***Table 4** — learned ranking and evidence (n = 290)*

ranking	r@10	Δ vs BM25	W/L	p
BM25	0.275	—	—	—
learned ranker, 16 features	0.332	+0.057 (0.019, 0.096)	31/14	0.016
+ evidence, 5 slots	**0.361**	**+0.086 (0.046, 0.126)**	**39/13**	**<0.001**

The learned combination is worth 5.7 points on repositories it has never seen and the evidence override 2.9 (12 wins, 3 losses, p = 0.035); removing the four graph features costs 1.7 with an interval through zero (12 wins, 8 losses, p = 0.50). Both readings are what a larger sample overturns.

***Table 5** — the ranker split by what the issue offers, at both scales (left 290 tasks, right 2,232)*

subset	n	BM25	ranker	n	BM25	ranker
quotes the gold body	54	0.481	**0.776**	355	0.459	**0.755**
names a path or a frame	16	0.406	0.469	224	0.380	**0.659**
neither	220	0.215	0.214	1,653	0.185	0.236

On the tasks whose issue pastes a line from the gold body the ranker gains **29.4 points at 290 tasks and 29.6 at 2,232** — a finding that does not care about sample size. The bucket 290 tasks badly misjudged is the middle one: naming a file or printing a frame looked worth 6 points on sixteen tasks and is worth **27.8** on 224. What the issue *states* is the signal; the graph is not.

The same claims on 2,232 tasks. Table 6 repeats them on the full test split — 2,174 snapshots, the same twelve repositories, the same code and denominators.

***Table 6** — what survives a 7.7× larger sample*

reading	290 tasks	2,232 tasks
file / symbol recall@10	0.624 / 0.275 = 2.27×	0.564 / 0.248 = **2.28×**
best graph arm, symbol	+0.012 (−0.026, 0.050)	+0.022 (0.013, 0.031)
best graph arm, file	−0.010	+0.000 (−0.009, 0.008)
learned ranker, symbol	+0.057	+0.113 (0.098, 0.128)
learned ranker, file	—	0.564 → 0.643
ranker without the graph features	−0.017, p = 0.50	+0.027 (0.017, 0.036)
evidence override over the ranker	+0.029, p = 0.035	−0.002 (−0.008, 0.004)

Three things move and one does not. The **gap is invariant**, 2.27× against 2.28×, which is what a property of ten slots and a file's definition count should do. The **graph is worth two points at symbol level and zero at file level**: at 290 tasks the first was invisible; at 2,232 it is +0.022 with an interval excluding zero, 169 wins to 90, positive in 8 of 12 repositories — real, and small, because it reorders symbols inside files text already found, while its file-level exchange is 81 wins to 94 losses. The **evidence override does not survive**: with 7.6× the training data the ranker reaches 0.361 alone and the override subtracts 0.2 points at five slots, 2.5 at ten, adds nothing at one. The 2.9 points it appeared to add were the four features it duplicates, finally able to learn.

How many tasks a claim needs. That a 2.9-point rule evaporated is arithmetic, not luck. Resampling the 2,232 paired differences and running the same sign test — 400 draws per cell — gives the share of n-task studies that would call an effect real: the ranker's +11.3 points is detected at n = 100 in 88% of draws, the graph's +2.2 in 39% at 290 and 87% at 800, the evidence rule's −0.2 never, at 1–4%. A 300-task comparison sees five-point effects or larger.

***Table 7** — share of resamples whose sign test reaches p < 0.05, by sample size*

effect, full scale	n = 100	290	500	800	2,232
ranker, +11.3 points	0.88	1.00	1.00	1.00	1.00
graph, +2.2 points	0.09	0.39	0.62	0.87	1.00
evidence, −0.2 points	0.01	0.02	0.03	0.04	0.00

Reproducibility. Each number here is re-derived from the released JSONL by a 220-assertion check script, so a claim that drifts from the data fails the build rather than the review.

6. Discussion

The practical claim is a reprioritisation. One-hop expansion is not where the error is: 88.1% of gold symbols are more than one hop from a strong text seed. The margin is in the source an issue quotes, and in candidate generation for the 27.6% of tasks whose gold never enters a top-500 pool.

The measurement claim is about granularity: file-level numbers near 90% are quoted as evidence that the hard part is over, yet the same ranking is 62.4% on files and 27.5% on symbols.

One thing would not overturn it: a stronger reranker over the same pool, since a perfect one reaches 0.600 and the graph's entire headroom is three points. What would overturn it is a different pool — a retriever that draws candidates by meaning rather than by name — or a graph built by type inference, since PyCG reports 99.2% precision at 69.9% recall [4].

7. Limitations

Python only, as are most benchmark tasks. The eight-arm comparison uses one retriever, so the graph result is about its marginal value *given this ranking*; the learned ranker is trained on six repositories and tested on six, which bounds overfitting but leaves 290 tasks for the grid, re-checked on 2,232 for the three headline readings. The evidence channel is bounded by what an issue states: on the 74% that name no source it does nothing, and pytest (sixteen tasks) loses 12.5 points to tracebacks that point at test files. We measure retrieval, not resolution: a better shortlist is necessary for a better patch, not sufficient.

8. Conclusion

Same ranking, two granularities: 62.4% of gold files, 27.5% of gold symbols. Eight graph arms, none of which improves symbol-level recall and three of which significantly hurt file-level recall; on 2,232 tasks the best of them is worth two symbol-level points and zero file-level ones. The arithmetic explains why: an 11.9-point reach ceiling, a ball that is 0.42% gold against 3.1% for a text slot, and a graph headroom of three points even with perfect reranking. The same ten slots reach **36.1%** once a ranker reads what the issue states — which is the difference between inferring structure and reading it.

References

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